

1 CORE 200 Reading | Module 1 | Causation: Preliminaries

1.0.1 Module 1: Causation | Overview

We will develop **six basic concepts**, all of which are prerequisite to a theory of causation:

1. observation vs. intervention: the difference
2. observation vs. intervention: predicting depends upon which you did and upon what causes what,
3. events vs. kinds of events,
4. particular cases of causation vs. causal generalizations,
5. the idea of a variable, and
6. the idea of a data table.

Causation is best understood as how the world responds to an intervention. We might **observe** that boys who watch the most violence on TV are also the boys who get into the most fist fights at school. This is the same as saying watching violent TV and having fistfights are **associated**. To further claim that the relationship is causal is to assert that if we were to **intervene** and curtail the violence these boys watch on TV, then other things equal the result would be fewer fistfights at school. TV watching and fist fights might be associated because watching TV causes fist fights, but it also might be that boys who are disposed to be violent like violent TV more, and in that case **intervening** to curtail their TV watching habits would not reduce their fist fights at all. The easiest way to differentiate between the two explanations of the association between TV violence and fist fights requires analyzing how things respond to an intervention.

✿ Example (A) Association vs. Causation: Eye Color

Right eye and left eye color are highly associated, but neither is the cause of the other. To say they are highly associated means that if you were to observe that someone's left eye was blue, you would be in good position to say that their right eye was also blue. If we were to intervene to surgically replace a random person's left eye with a blue one, however, their right eye would not change color to blue. This is because the genes you inherited from your parents are a common cause of the color of both of your eyes.

✿ Example (B) Association vs. Causation: Winter Coats and Air Temperature

The proportion of people on the street wearing a heavy winter coat is highly associated (negatively) with the outside air temperature. Does wearing a winter coat cause the air temperature to be low? Obviously the causation is the other way – low air temperature causes people to wear coats. To make that clear to a visitor from another planet, however, use the idea of an intervention: If we intervened to make everybody wear a heavy winter coat – the air temperature would not change. If we intervened to lower the air temperature dramatically, however, the proportion of people wearing a winter coat would change.

Particular cases of causation, like identifying what caused the space shuttle Challenger to blow up just a few minutes after its launch in 1986, are usually the focus of legal proceedings or hearings and not research studies, thus they are not our focus. We are instead targeting causal generalizations, like whether spending more money on public school teacher salaries improves the education their students receive.

Causal generalizations involve **variables**, for example weight, income, or the number of violent crimes per capita per year. Variables are the foundation of data collection and statistical analyses, and therefore we must attend to them.

1.1 Observing vs. Intervening

1.1.1 Module 1 | Seeing vs. Doing

Perhaps the most fundamental concept in causal reasoning is the difference between observing and intervening. By observing or seeing we mean passively observing something. By intervening or doing we mean actively changing something. For example, you might observe that your garden is muddy, or you might intervene to make it muddy by spraying it with a hose.

Sometimes passive observation involves acquiring data through measurement or other actions: weighing, asking a question, and so on. As long as actions of this sort do not change the thing that is the focus of interest, say the person's

weight, it is still passive observation. The “passive” in passive observation, then, does not necessarily indicate a complete lack of action on the part of the observer. Rather it means that whatever actions the observer takes do not change the thing being observed in any relevant way.

✿ **Example (C) Observation vs Intervention: Paper length, Rulers, and Paper Cutters**

Measuring the length of a piece of paper with a ruler is a passive observation. It will not make the paper longer or shorter. Cutting the paper with a machine to a length of 5 inches is an intervention.

✿ **Example (D) Observation vs Intervention: Income**

Examining someone’s most recent pay stub and finding their income is \$4,000 is a passive observation of their monthly income. It will not make their income higher or lower. Giving (or taking) enough money so that their monthly income is exactly \$4,000 is an intervention.

Sometimes the mere presence of an observer can change the thing being observed. For example, the behavior of school children may be very different when their teacher is watching than when they are left unattended. In such circumstances, passive observation might require such devices as one-way mirrors, hidden cameras, and so on. However, in most of the cases that will concern us, passive observation does not necessitate complications of this kind. Hence, whenever you come across an example in which data is gathered solely by watching, asking questions, or measuring, you can assume that this is a case of seeing unless there is some clear reason to suppose otherwise.

✿ **Example (E) Observation vs Intervention: Income**



Imagine that you have a backyard with a garden. On Monday, you look out your window and see that your garden is wet: water is dripping from the leaves and puddles in the soil. You have observed (seen) that the garden is wet. On Wednesday, you go into your garage, turn on a sprinkler, and water the garden for 1/2 hour until there is water dripping from the leaves and puddles in the soil. You have done something to the garden; you have intervened to make it wet.

The distinction between seeing and doing is fundamental to causal and statistical reasoning. Words like assigning, manipulating, and intervening indicate doing rather than merely seeing. Whenever you see one of these words, you should think of it as a red flag going up, signaling that what you are dealing with is an example of doing, or intervening. Words like observing, passively observing, recording, surveying and measuring indicate seeing, or passive observation.

1.1.2 Module 1 | Predicting, Seeing, and Doing

The conclusions you can draw from cases in which you have seen something are not always the same as those you can draw cases in which you have done something. If you believe that one thing like exercise causes another like weight loss, for example, then you believe that intervening on the cause (starting an exercise regime) should result in a change in the effect (you should start to lose weight). If two things are merely associated, however, like having tar-stained fingers and getting lung cancer, then intervening on one will not necessarily produce a change in the other. To reinforce this point, which is absolutely central to the idea of causation, consider again the garden example.

✿ **Example (F) Predicting From an Observation vs. Predicting from an Intervention: Gardens and Sprinklers Again**

On Monday, after seeing that your garden is wet, you might say to yourself, “it must have rained, in which case my neighbor’s garden must be wet too.” Odds are you’re right. On Wednesday, having just made your garden wet with a sprinkler, you wouldn’t infer that it has just rained, or that your neighbor’s garden was wet either. The doing case is not quite so simple, because what you can predict depends on what you observed before you turned on the sprinkler and did something. Suppose first that you turned on the sprinkler on Wednesday afternoon after having looked at your garden and seen that it was dry. In that case you could infer that your neighbors garden was probably dry too, unless she too had decided to water. This is a really a case of seeing and then doing. Suppose, alternatively, that on Tuesday night you set the sprinkler to go on automatically 1/2 hour before you were to wake up on Wednesday morning. Suppose you wake up Wednesday morning, look out your window (which only gives a view of your garden), and see that it is wet. This is really a case of doing without seeing. Now you say to yourself, “Hmm, I wonder if my neighbor’s garden is wet?” Maybe it rained while you slept, in which case her garden would be wet too. But on the other hand, maybe it didn’t rain; you just don’t know.

There are three different cases:

- Just Seeing
- Seeing and then Doing
- Just Doing

After just seeing that your garden was wet you could confidently predict that your neighbor's garden would be wet. After seeing that your garden was dry and then watering it, you could predict that your neighbor's garden would be dry as well (but less confidently as she might have watered it). After making your garden wet without looking by having set the sprinkler the night before, you can't make any prediction about the state of your neighbor's garden.

1.1.3 Module 1 | Prediction and Causation

What you can predict depends not only on whether you observed or intervened, but also on what is causing what. If you are predicting one thing A from either just seeing or seeing and then doing another thing B, then it doesn't matter whether A causes B or vice versa, or even whether something else is causing both A and B. In all these cases, learning about B *will* help you predict A. If you are asked to predict A after just intervening on B, however, it *does* matter how A and B are causally connected. If A is a cause of B or there is a common cause, then you won't get any help in predicting A after just doing an intervention on B. If B is a cause of A, however, then you will be able to predict A after just doing an intervention on B.

✿ Example (G) Predicting Depends on the Causal Structure: Smoking, Tar-Stains, and Dentures

Consider again smoking and tar stains, this time on dentures.



Suppose you see two 60 year old men with dentures, one who has serious tar-stains on his dentures and one who does not. You might infer that the one with stained dentures smokes a lot and predict that he has a higher chance of getting lung cancer than his friend. Suppose however, that you were then shown two 60-year-old women who were, just before you saw them, given previously owned dentures to wear for the next year. One was given tar stained dentures and one unstained dentures. Since tar-stains on dentures don't really cause lung cancer, you can't use the information about who has stained dentures to tell you who has a higher chance of getting lung cancer.

Why are the cases of smoking and income different? Tar-stained dentures *do not* cause lung cancer, but higher incomes *do* cause higher taxes. We will return to this theme repeatedly. It is the key to the subject.

1.1.4 Module 1 | Particular Events and Causal Generalizations

In this section we distinguish between particular causal claims and causal generalizations. Particular causal claims involve an event, for example the terrorist attacks on the World Trade Center of September 11th, 2001, while causal

generalizations involve policy, like whether tax cuts stimulate the economy or cause deficits which drag down the economy. Consider two examples.

✿ **Example (I) Particular Causal Claims: Mr. Host and Mr. Guest**

Imagine two lawyers are arguing a lawsuit in which a guest's child was accidentally shot by the host's child at a party. Mr. Host was showing Mr. Guest his new gun, when he was called away to put out a fire in the kitchen. Mr. Guest put away the gun, but later their two sons got into the gun cabinet and played with the gun. Mr. Host's son tripped, and accidentally shot Mr. Guest's son in the leg.

- "The defendant, Mr. Host, was negligent, because he left the gun loaded."
- "My client, Mr. Host, is not negligent. He left the study where the gun was kept to tend to an emergency in the kitchen, and he instructed Mr. Guest to put the gun away and pull the cabinet door shut to lock it. He had no reason to suspect that Mr. Guest's son knew where the key to the cabinet was hidden. If Mr. Guest hadn't irresponsibly told his son where Mr. Host hid his key, the boys would not have gotten into the gun cabinet, and this tragedy never would have happened!"
- "Your honor, Mr. Host was doubly negligent. If he had not removed the child-lock protection that came with the gun, then we would most certainly not be here today. "
- "I really must differ. Mr. Host's son was 13, and already knew how to disable a child-safety lock. The lock would have made no difference."

✿ **Example (J) Causal Generalizations: School Vouchers**

Imagine that two political candidates are debating the merits of public school vouchers. The Republican candidate supports school vouchers, while the Democratic candidate does not.

- Republican: "Giving families vouchers will lead to more competition among schools, and more competition among schools will lead to better schools. Experiments in Milwaukee have already demonstrated that students who take advantage of vouchers learn more than those who wanted to use vouchers but could not. Our entire society is predicated upon the power of the free market. We don't allow monopolies in this country because they hurt the consumer. Why should we allow the public schools to have a monopoly on education?"
- Democrat: "My opponent makes things sound so simple – would that it were so. First, there is no solid statistical evidence that vouchers have any positive effect on students who use them. Second, vouchers will never provide enough money to fully fund a private education. Third, vouchers take precious money away from the public schools, which are already under funded. The children left behind who can't afford to supplement the voucher to go to a private school will be dramatically worse off than they are now. The effect of this policy will be to ruin the public educational system, a system that, while not perfect, is the primary means by which our society can remain egalitarian. Vouchers do not provide equal opportunity for all, and without that we are lost. I propose we improve the public schools – not gut them."

Both of these examples involve causation, but in very different ways. The legal dispute involves a single event: an accidental shooting at a particular party. It seeks to identify which of many possible actions or non-actions caused the accident, and which of these, if any, are the responsibility of the defendant. The school voucher policy dispute is not about a single event, but rather how society in general will respond to an intervention that affects thousands of people. Both candidates are well aware that some students will benefit and some suffer under either policy. The argument is not about a single case but rather about general trends.

Reasoning about particular causal claims depends upon our beliefs about causal generalizations. For example, suppose your friend Mark gets the flu a few days after going to a party on Monday. He claims that he got the flu on Wednesday because he went out in the rain on Tuesday. You tell him, "no, you got the flu because you shook hands with Dave at the party on Monday, who still isn't over his flu." Regardless of who is right, your explanation depends upon believing that, in general, shaking hands with people who have the flu causes getting the flu. Mark's account depends on his believing that, in general, going out in the rain causes getting the flu. A lawyer who argues that his client was stricken with lung cancer because he smoked for 50 years believes that, in general, smoking causes lung cancer, and that his client was an instance of this generalization.

Although reasoning about causation in particular cases is fascinating and a topic unto itself, our focus will be on causal generalizations. We want to make sense of what we *mean* when we make claims like, in general, exposure to high levels of radiation causes leukemia, and we want to understand the scientific connection between causal generalizations and evidence, especially statistical evidence.

1.1.5 Particular Events and Causal Generalizations

✿ Example (K) Particular Causal Claim: The Space Shuttle Columbia

On February 1st, 2003, the space shuttle Columbia broke up over Texas during re-entry, killing seven astronauts. In a report finalized many months later, NASA chief Sean O’Keefe admitted his agency “just flat missed” what investigators named as the accident’s cause – foam that hit the shuttle’s wing shortly after liftoff.

✿ Example (L) Causal Generalization: Cholesterol and Carbohydrates

In the early 21st century, several new studies began to support the previously controversial diet theories of Dr. Atkins. According to Atkins, eating simple carbohydrates like white bread or pasta produce a blood-sugar spike which in turn causes the body to release insulin quickly, which causes the body to store fat and to produce LDL – the harmful low density cholesterol indicated as a risk factor in heart disease.

1.2 Events, Kinds of Events, and Variables

1.2.1 Module 1 | Events and Conditions

In a causal generalization, one thing is said to generally cause another. What are these *things* that are related as cause and effect, however? In our account, which focusses on connecting causal claims to statistical evidence, the answer is that they are *variables*. To get to variables, however, we will begin with particular events and then group events of a kind together in order to arrive at variables that take on a particular range of values that can then be measured and recorded in a statistical data table. In this section we discuss events.

Consider a few examples of events in which one thing caused another:

1. A cue ball hits the eight ball and causes the eight ball to fall into the corner pocket.
2. My daughter knocks over her glass of milk and it soaks my new pants.
3. The Titanic hits an iceberg and then sinks.
4. Pushing the gas pedal causes the car to accelerate.

These are all cases in which one *event* produces, or is responsible for, another *event*.

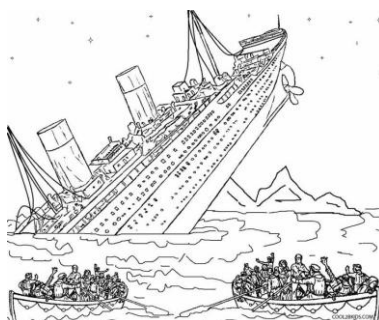
Besides events, we also identify *conditions* as causes or effects. For example, following are more naturally viewed as conditions than events:

1. The air temperature being below 32 degrees Fahrenheit is a condition that causes the water on the lake’s surface to turn to ice.
2. A street having a steep slope is a condition that causes a ball to roll down to the bottom of the hill.
3. The low density of oxygen at the top of Mt. Everest is a condition that causes even moderate exercise to be exhausting.

Conditions are enduring features of a situation, while events are changes in a situation. We group events and conditions, and call them “factors.”

The distinction between events and conditions is not nearly as clean as we have made it out to be. Which of the possible causal factors in the car simulation above is best described as an “event?” Which are best described as a “condition?” Perhaps *any* of the causal factors in the car may be described either as conditions or events. We can describe the event “pushing on the accelerator” as the condition “the accelerator is fully depressed.” We can describe the condition “lights are on” as the event “the lights are turned on” we say that a causal factor is a condition if it is a relatively stable or enduring feature of the situation, e.g. the gas tank being full. A causal factor is called an event if it is an activity that changes a feature of the situation, e.g. setting the parking brake.

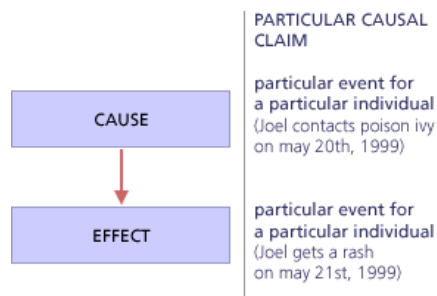
1.2.2 Module 1 | Events to Kinds of Events



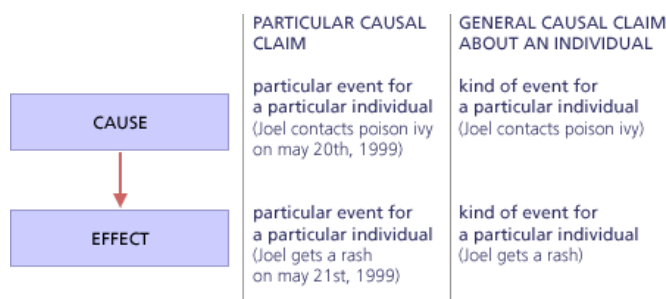
The sinking of the Titanic was a single event, and one of its causes was another single event: hitting an iceberg. (This is a bit of an idealization. Sinking is not an instantaneous event, nor does it occur at a single point in space). Likewise, when we say: “The live Polio vaccine given to Doug Jones on May 1st, 1955 at noon caused a Polio infection which killed him,” we mean that the particular vaccine administration event was responsible for that case of Polio. Each of these examples involves two particular events: one for the cause and one for the effect. We want more than particular claims concerning one cause event and one effect event, however. We want to generalize to how vaccine administration events cause Polio or its absence in children overall.

We generalize from particular causal events to general causal claims in two steps. First we generalize from a particular cause and effect event involving a single individual to *events of a kind* involving a single individual. Then we move from single individuals to a *group of individuals*.

Consider a single cause event: Joel Smith rubs against poison ivy as he is walking through the woods on May 20th, 1999, and a single effect condition: Smith has an itchy rash that starts the next day and lasts a week.



The fundamental strategy for generalizing is to categorize separate events as being the same *kind of event*. For example, Smith might have rubbed against different poison ivy plants at different times. Although each event is separate, they are *events of a kind*: Smith rubbing against poison ivy. Similarly, the rash that appeared on Smith the evening of May 21st, 1999 was one event, but rash appearances like it on Smith are events of a kind for Smith. The individual remains the same, Smith, but similar events across different times can be lumped together as events of a kind. In so doing, we can make a general causal claim about an individual: “Whenever Joel Smith contacts poison ivy, it causes him to get a rash soon thereafter.”



A single event may be categorized as many different kinds of events. For example, the particular event: Joel Smith rubbed against poison ivy as he was walking through the woods on May 20th, can be categorized as: Smith rubbed against poison ivy, or Smith rubbed against a plant, or Smith walked through the woods.

The next move is to generalize from a single individual like Smith to a group of individuals, like adult males. Contacting poison ivy is a kind of event that any adult male can experience at any time, as is getting an itchy rash. What policy

really requires, however, is a causal claim about a population (a group): “Contact with poison causes itchy rashes among adult males.”

	PARTICULAR CAUSAL CLAIM	GENERAL CAUSAL CLAIM ABOUT AN INDIVIDUAL	GENERAL CAUSAL CLAIM ABOUT A POPULATION
CAUSE	particular event for a particular individual (Joel contacts poison ivy on May 20th, 1999)	kind of event for a particular individual (Joel contacts poison ivy)	kind of event for a population of individuals (an adult male contacts poison ivy)
EFFECT	particular event for a particular individual (Joel gets a rash on May 21st, 1999)	kind of event for a particular individual (Joel gets a rash)	kind of event for a population of individuals (an adult male gets a rash)

So there are two steps in moving from particular events involving causation to causal generalizations:

1. A *particular event* that occurs to a particular individual → a *kind of event* that occurs to a particular individual at many times.
2. A kind of event that occurs to a *particular individual* at many times → a kind of event that occurs to many *different individuals* at many times.

In the table below, each row has a particular event, and then a kind of event for that individual of which the particular event is an instance, and finally the kind of event generally. We have left parts of two rows blank, and ask that you complete them in the exercises below.

Table 1: PARTICULAR EVENTS AND KINDS OF EVENTS

Particular Event	Kind of Event for an Individual	Kind of Event Generally
Scheines climbs Giant Mountain in the Adirondacks on August 8th, 1999	Scheines climbing a mountain	Climbing a mountain
Jones watches the Steelers play the Cowboys on September 7th, 1997	Jones watches a football game	Watching a football game
The Exxon Valdez has an oil spill in Alaska in 1982	An oil spill for the Exxon Valdez	An oil spill
Bill Clinton's campaign for political office in 1992		
The U.S. Stock Market crash of October 17th, 1987		
Glymour purchasing a new car in August, 2003	Glymour purchases a new car	A new car purchase

1.2.3 Module 1 | Variables and Data Tables

Grouping events as events of a kind, like getting a flu shot, or grouping conditions as conditions of a kind, like being Latino, is the fundamental move to create *variables*, the main objects of statistical analysis.

When scientists collect data for statistical analysis, they don't write down events or conditions, they typically construct a *data table* in which each column is a *variable* and each row an *individual*. For example, in studying smoking and lung condition, they might collect data on four individuals and store the data in the following table:

Individual	# of Cigarettes Per Day	Lung Condition
Smith	5	Good
Jones	25	Fair
Gonzales	10	Good
Wright	30	Poor

Connecting this sort of statistical data table to general causal claims is, in some sense, the essence of causal and statistical reasoning.

In statistical data tables, each row refers to a particular individual in some *population*. The population need not be people; it could be countries, cities, dogs, trees, or families, etc. Whether it is people or not, however, it should be a definite group. For example, all adult U.S. citizens is a population and all countries with any territory north of the equator is a population, but “good looking people” is not. The data table that is actually analyzed rarely includes every individual in a population, however. The set of individuals the table does include is called a *sample* from the population. For example, if CNN declares that 48% of U.S. men favor tax cuts, they are describing a population that has over 100 million members. They usually go on to say that their numbers come from a random sample of about 1,000 men, which makes their overall results only accurate to within 3 or 4 %. Their data table had about 1,000 rows, and at least one column with a variable that indicates the individual’s preferences about tax cuts.

The columns in the table are variables: that is, quantities that can take on different *values* for different individuals. The connection between events, conditions, kinds of events and variables is this: *kinds of events (conditions)* can be represented as *values of a variable*.

Variables have a range of possible values. For example, Hair Color is a variable with possible values of red, blonde, brunette, etc. Something as simple as a person’s sex is a variable: it has two possible values: male and female. Some variables have only two possible values in no particular order, like Sex. Others have a small number of values that have a natural order, like Drink Size, with values = small, medium, or large, and others have an infinity of values, like Height in inches. Variables ought to have a range of values that correspond to a natural breakdown of the kinds of events or conditions that might obtain and that might make a difference.

For example, consider three kinds of events that relate to Polio vaccines:

1. Had “live” vaccine [live]
2. Had “dead” vaccine [dead]
3. Had no vaccine [none]

We might create a variable called Vaccine Received that can take on exactly one of these values, which we coded in the list above with the word in the square brackets, for example, [live]. An example of a simple data table involving five individuals in which we record the kind of Polio vaccine each received is as follows.

Individual	Variable: Vaccine Received
Lopez	live
Mitchell	live
Shaffer	none
Goodstein	dead
Sufez	dead

What we really care about, of course, is connecting the kind of vaccine the child received with their health outcome, so a better table records two variables for each individual: Vaccine Received and Health Outcome. What values should Health Outcome range over? Perhaps [OK] and [Polio]. Then a table that recorded Vaccine Received and Health Outcome might look as follows.

Individual	Variable: Vaccine Received	Variable: Health Outcome
Lopez	live	OK
Mitchell	live	Polio
Shaffer	none	OK
Goodstein	dead	OK
Sufez	dead	OK

Lots of different quantities can be represented as variables. Lots of different kinds of events, or kinds of conditions, can be represented as a value of a variable. A person’s age is a variable. The average amount of vitamin C a person consumes a day is a variable. The number of people who live in a city is a variable.

With data tables like the ones above, scientists aim to see if associations exist between events of one type (getting the live Polio vaccine), and events of another (not having Polio).

Enduring conditions can be treated in the same way. In the case of the Apollo 1 fire, for example, we might construct the following table to show the changes instituted between Apollo 1 and Apollo 2. The concentration of oxygen is a condition, not an event, yet it can easily be represented as the value of a variable.

Individual	Electrical Short	Oxygen Concentration	Fire
Apollo 1	True	100%	True
Apollo 2	False	18%	False

1.2.4 Module 1 | Practice with Variables and Data Tables

A key skill in critically assessing media reports on causal studies is identifying the variables and individuals under study. If you can reconstruct at least the sort of data table that was measured and analyzed, you will understand a lot about the study. In this section we present an example in which we identify the variables and kind of individuals covered, and then ask you to read a few case studies and do the same for each of them: identify the variables and kind of individuals discussed.

Read the following study investigating the relationship between being exposed to Agent Orange and having a child with leukemia. Ask the following questions as you read. What are the main variables in this study? What values does each variable range over?

Example (M) Possible link between Agent Orange, leukemia

The main variables concern having leukemia, having a father who served in either Vietnam or Cambodia, and having a father who was exposed to Agent Orange. The variables, along with their values in brackets, are plausibly as follows:

- Has Leukemia: [yes, no]
- Father Served (in Vietnam or Cambodia): [yes, no]
- Father Exposed (to Agent Orange): [yes, no]

What type of individuals would be recorded in each row of the data table? Would there be a row for each family analyzed? No, because the variable Has Leukemia doesn't apply to a family, but rather to a child. The individuals are children – some healthy and some with leukemia. So the first few rows of a data table for one of these studies might look as follows:

Individual	Has Leukemia	Father Served	Father Exposed
Jordan	Yes	Yes	No
Jones	No	No	No
Gonzales	No	Yes	No
Underwood	Yes	Yes	Yes

Each of the variables above ranges over only two values. In other cases, like weight, a variable's range might be continuous within a certain range – say 0 to 1,000 pounds. We notate a range of values with dots, like so: Weight [0..1,000]. We won't split hairs over whether a variable is continuous or not. For example, we still might notate the range of values for a variable like Age in Years, which is technically discrete as so: Age in Years [0..100].